

WAIFINDERS — FULL BUILD METRICS

Complete Technical State · All Modules · All Validation Numbers

Generated: 2026-08-09 · For ChatGPT Context Update

8	Active WARN domains 260	Tests passing (all repos) 58	BC Nations scored 67	Nuclear plants tracked 2,000
0.8818	ROC-AUC WARN Alberta (real data) 0.2837	PR-AUC WARN Alberta 7.15×	Top-10 Lift Over random baseline 274 h	Mean lead time Alberta fires 5

A. PLATFORM ARCHITECTURE

Engine	Abbreviation	Function	TRL	Status	Output file
Wildfire / Multi-hazard Warning	WARN	0-1 composite score per hazard; 4-tier operational state	1	PILOT-READY	warn_*.json (8 files)
Infrastructure Stress	PULSE	Asset condition + compound hazard zone overlay	1 → 2	PILOT-READY	LOWESS_bands.json
Multi-hazard Decision	WISE	Synthesises all WARN domains; compound escalation	1	PILOT-READY	wise_multihazard_decision.json
Audit / Evidence	SENTINEL / S.A.F.E.	BLAKE3-hashed evidence chain; federal-grade audit trail	5	PILOT-READY	BLAKE3 manifests
Indigenous Readiness	SPARK7	10-year readiness projection: pressure, readiness	10, 10	PILOT-READY (BC connected)	spark7_bc_nations.json

Hashing standard: BLAKE3 only. No SHA-256, no MD5 in any WAIFINDERS output.

B. WARN ENGINE — ALL 8 ACTIVE DOMAINS

Formula pattern: warn_score $\in$ [0,1]. Tiers: 0=NORMAL, 1=MONITOR, 2=MITIGATION, 3=EMERGENCY.

Domain	Formula	Thresholds	Data source	Top result	Output
Wildfire Alberta	$\text{FWI_norm} \times 0.40 + \text{spread_norm} \times 0.25 + \text{pop_norm} \times 0.20 + \text{resource_norm} \times 0.15$	$E \geq 0.75, M \geq 0.55, MO \geq 0.30$	Open Alberta OGL-A 2006–2025	Fort McMurray 2016: 0.89 EMERGENCY	warn_ab_wildfire_events.json
Wildfire BC Nations	$\text{FWI_norm} \times 0.35 + \text{proximity_norm} \times 0.30 + \text{area_norm} \times 0.20 + \text{spread_norm} \times 0.15$	$E \geq 0.70, M \geq 0.45, MO \geq 0.20$	BC Wildfire Service anchor fires 2017–2023	Kwadacha Nation: 3.7246 EMERGENCY	warn_bc_wildfire_nations.json
Earthquake	$\text{mag_norm} \times 0.45 + \text{shallow_norm} \times 0.35 + \text{pop_norm} \times 0.20$	$E \geq 0.70, M \geq 0.45, MO \geq 0.20$	USGS ShakeMap / ISC-GEM	Cascadia M9.0: 0.748 MITIGATION	warn_earthquake_events.json
Tsunami	$\text{wave_norm} \times 0.45 + \text{mag_norm} \times 0.35 + \text{reach_norm} \times 0.20$	$E \geq 0.70, M \geq 0.45, MO \geq 0.20$	NOAA DART buoy network	Tohoku 2011: 0.970 EMERGENCY	warn_tsunami_events.json
Flood Surge	$\text{surge_norm} \times 0.60 + \text{precip_norm} \times 0.40$	$E \geq 0.70, M \geq 0.45, MO \geq 0.20$	USGS NWIS / NOAA CO-OPS	Sandy 2.74m: 0.549 MITIGATION	warn_flood_surge_events.json

Domain	Formula	Thresholds	Data source	Top result	Output
Hurricane	wind_norm×0.45 + surge_norm×0.35 + proximity_norm×0.20	E≥0.70, M≥0.45, MO≥0.20	NOAA NHC HURDAT2 1851–2023	Andrew 1992: 0.814 EMERGENCY	warn_hurricane _events.json
Nuclear	capacity_norm×0.45 + epz_pop_norm×0.35 + power_norm×0.20	E≥0.65, M≥0.40, MO≥0.20	NRC Power Reactor Status (daily)	Indian Point: 0.871 EMERGENCY	warn_nuclear _plants.json
Nuclear (formula components)	capacity_norm = MWe/1299 epz_pop_norm = pop/350000 power_norm = pct/100	Conservative EMERG threshold ≥0.65	67 plants. 51/67 matched to NRC	Limerick: 0.771 Seabrook: 0.733	67 plants scored

C. WISE MULTI-HAZARD DECISION ENGINE

```
base_tier = max(tier_int for all domains) # 0=NORMAL ... 3=EMERGENCY if count(domains with tier >= MONITOR) >= 2:
base_tier = min(3, base_tier + 1) wise_decision = tier_name[base_tier]
```

Domain	Score	Tier
Wildfire	0.0	NORMAL_OPERATION
Earthquake	0.399	MONITOR
Tsunami	0.97	EMERGENCY_RESPONSE
Flood Surge	0.549	MITIGATION_REQUIRED
Hurricane	0.814	EMERGENCY_RESPONSE
Nuclear	0.871	EMERGENCY_RESPONSE

Field	Value
wise_decision	EMERGENCY_RESPONSE
compound_event	True
elevated_hazards	['earthquake', 'tsunami', 'flood_surge', 'hurricane', 'nuclear', 'pulse']
generated_utc	2026-08-09T07:24:52
Output file	wise_multihazard_decision.json (1,995 bytes)
TRL	5 (upgraded from 3 in Aug 2026 sprint)

D. PULSE — INFRASTRUCTURE STRESS

```
pulse_base = 0.65 × poor_condition_risk + 0.35 × age_norm compound = min(1.0, pulse_base × (1 + 0.25×seismic_HIGH
+ 0.20×tsunami_zone + 0.15×flood_SFHA)) bands: RED≥0.70 AMBER≥0.45 YELLOW≥0.25 GREEN<0.25
```

Metric	Value
Bridge count	2000
RED band	200
AMBER band	340
YELLOW band	1141
GREEN band	319
Seismic source	USGS NSHM 2014 (E2014R1) — 2%/50yr PGA at county centroid
Seismic zone NYC	HIGH (PGA 0.17–0.18g) — Kings, Queens, Richmond, Bronx, New York
Tsunami source	NOAA CO-OPS active tide gauge presence + coastal geography
Flood source	FEMA NFHL county-level SFHA Zone A/AE — 11/13 NY counties confirmed
Toronto AUC	0.7384 (n=4,248, temporal holdout, OGL-T)

Metric	Value
Calgary AUC	0.6626 (n=4,748, temporal holdout, OGL-Calgary)
GATE-004	CLOSED — Toronto, Calgary, Kitchener provenance confirmed
Output file	pulse_disaster_exposure.json (5,999 bytes)

E. WARN NUCLEAR — DETAIL

Plant	State	MWe	EPZ Pop	Power %	Score	Tier
Indian Point	NY	1041	0	100%	0.871	EMERGENC
Limerick	PA	1134	0	100%	0.771	EMERGENC
Seabrook	NH	1246	0	100%	0.733	EMERGENC
Fermi	MI	1122	0	100%	0.725	EMERGENC
Peach Bottom	PA	1112	0	100%	0.697	EMERGENC
San Onofre	CA	1127	0	100%	0.694	EMERGENC
Catawba	SC	1129	0	100%	0.679	EMERGENC
McGuire	NC	1100	0	100%	0.678	EMERGENC
Susquehanna	PA	1257	0	99%	0.676	EMERGENC
Millstone	CT	1227	0	100%	0.671	EMERGENC
Zion	IL	1040	0	100%	0.668	EMERGENC
Nine Mile Point	NY	1299	0	100%	0.668	EMERGENC
Perry	OH	1261	0	96%	0.660	EMERGENC
Grand Gulf	MS	1297	0	100%	0.658	EMERGENC
Vogtle	GA	1254	0	100%	0.649	MITIGATI

Metric	Value
Plants scored	67
NRC units matched	51
NRC report date	8/7/2026 12:00:00 AM
Output file	warn_nuclear_plants.json (23,202 bytes)
BLAKE3	20e748a2b77257a6...

F. WARN BC WILDFIRE NATIONS + SPARK7

Metric	Value
Nations scored	58
EMERGENCY_RESPONSE	1
MITIGATION_REQUIRED	57
MONITOR	0
Territory source	BC Data Catalogue — First Nations traditional territories (public, approximate centroids)
Fire data source	failed (HTTP Error 404: Not Found) — using anchor fires — using recent anchor fires (2021+)
Output file	warn_bc_wildfire_nations.json (24,393 bytes)
BLAKE3	323e97c451c4c9c0...
SPARK7 integration	warn_spark7_bridge.py — WARN score ×10 → SPARK7 wildfire pressure
SPARK7 BC nations	data/spark7/spark7_bc_nations.json (58 nations, GeoJSON)

Metric	Value					
SPARK7 band result	All 58 nations: ORANGE (readiness10 53–55, horizon 10yr, decay 0.3/yr)					
OCAP	Ownership Control Access Possession — no Nation-specific data collected					
Rank	Nation	Region	WARN Score	WARN Tier	Closest Fire	km
1	Kwadacha Nation	Rocky Mountain	0.725	EMERGEN	Donnie Creek Complex	44.0
2	Tsay Keh Dene Nation	Rocky Mountain	0.667	MITIGATI	Donnie Creek Complex	82.1
3	Lytton First Nation	Fraser Canyon	0.641	MITIGATI	Lytton Creek	1.1
4	Halfway River First Nation	Peace	0.61	MITIGATI	Donnie Creek Complex	120.3
5	Okanagan Indian Band	Okanagan	0.6	MITIGATI	White Rock Lake	21.4
6	Bonaparte Indian Band	Thompson	0.598	MITIGATI	Tremont Creek	25.6
7	Nlaka'pamux Nation Tribal Council	Thompson	0.595	MITIGATI	Lytton Creek	31.6
8	Lower Nicola Indian Band	Nicola	0.593	MITIGATI	Lytton Creek	33.0
9	Upper Nicola Band	Nicola	0.592	MITIGATI	White Rock Lake	26.5
10	Okanagan Nation Alliance	Okanagan	0.589	MITIGATI	White Rock Lake	29.1
11	Westbank First Nation	Okanagan	0.588	MITIGATI	White Rock Lake	29.6
12	Nooaitch Indian Band	Nicola	0.557	MITIGATI	Lytton Creek	57.2
13	St'át'imc Nation	St'át'imc	0.555	MITIGATI	Lytton Creek	58.7
14	Tk'emlúps te Secwépemc	Thompson	0.551	MITIGATI	Tremont Creek	57.1
15	Kamloops Indian Band	Thompson	0.547	MITIGATI	Tremont Creek	59.2
16	Coldwater Indian Band	Nicola	0.541	MITIGATI	Lytton Creek	67.9
17	Splatsin First Nation	Shuswap	0.539	MITIGATI	White Rock Lake	62.3
18	Canim Lake Band	Cariboo	0.536	MITIGATI	Horse Lake Complex	44.4
19	Little Shuswap Lake Band	Shuswap	0.53	MITIGATI	White Rock Lake	68.1
20	N'Quatqua	Sea to Sky	0.527	MITIGATI	Lytton Creek	76.8

G. WARN WILDFIRE VALIDATION — ALBERTA REAL DATA

Metric	Value	Notes
Dataset	Open Alberta OGL-A 2006–2025221e7a0-4f46-4be7-9c5a-e29de9a3447e	
Training events	20,848	Years 2006–2019
Test events	6,980	Years 2020–2025, temporal holdout
High-impact class	158 events	Large / extreme fires
ROC-AUC (test)	0.8818	REAL DATA — cite this only
PR-AUC (test)	0.2837	
Top-10 Lift	7.15×	
Top-20 Lift	4.08×	
Mean lead time	274.42 h	
Median lead time	17.75 h	
Formula features	14	FWI, temp, humidity, wind, precip, drought, FFMC, DMC, DC, ISI, BUI, DSR, spread, pop
SUPERSEDED	ROC-AUC 0.998	Label leakage on 500-event synthetic set — DO NOT CITE
Scope	Alberta only	Must NOT transfer to BC/Ontario without separate replay
Peer review	NOT COMPLETE	GATE-002 open

H. BC WILDFIRE SERVICE ANCHOR FIRES (public records)

Fire	Year	Area (ha)	Lat	Lon	Used for
Donnie Creek Complex	2023	589,552	57.0	-124.2	WARN proximity; SPARK7 pressure; Kwadacha Nation
Lytton Creek	2021	83,000	50.3	-121.6	Lytton FN proximity
White Rock Lake	2021	83,334	50.17	-119.6	Okanagan proximity
Tremont Creek	2021	22,035	51.0	-121.3	Interior FN proximity
Horse Lake	2021	18,874	51.5	-121.8	Cariboo FN proximity
Plateau Complex	2017	521,000	51.5	-124.0	Tsilhqot'in proximity
Elephant Hill	2017	190,000	50.8	-121.2	Thompson FN proximity
Hanceville-Riske Creek	2017	230,000	51.7	-123.0	Chilcotin FN proximity
Polychrome Complex	2022	62,000	59.5	-127.0	Northern BC proximity

I. GATE CLOSURE STATUS (waifinders_disasters)

Gate	Name	Status	Changed Aug 2026?	Blocker
GATE-001	Official Alberta Fire Data	REPLAY_COMPLETE REVIEW_PENDING	No	External dependency
GATE-002	External Peer Review	NOT_CLOSED	No	Recruit ≥3 reviewers
GATE-003	Production Claim Gate	PILOT_READY	No	Needs GATE-001+002
GATE-004	Water Validation Provenance	CLOSED	Confirmed	None
GATE-005	Bow River Digital Twin	NOT_CLOSED	No	GIS + Treaty 7
GATE-006	Federal SENTINEL	READY_TO_SEND	No	Outreach only
GATE-007	Deployment Ops	CLOSED	No	None
GATE-008	Doc Scrub	CLOSED	No	None
GATE-009	Tests + BLAKE3 Manifest	CLOSED	260 tests	None

J. TEST INVENTORY — 260/260 PASSING

Test file	Tests	Module tested	Key anchors
test_warn_nuclear.py	45	warn_nuclear_plants.py	Indian Point 0.871 EMERG, Palo Verde MITIGATION
test_warn_bc_wildfire_nations.py	51	build_warn_bc_wildfire_nations.py	Kwadacha EMERG, Lytton MITIGATION+, haversine
test_warn_spark7_bridge.py	22	warn_spark7_bridge.py	58 nations, WARN→SPARK7 pressure, OCAP provenance
test_warn_earthquake.py	~15	warn_earthquake_events.py	Cascadia M9 0.748, magnitude norm
test_warn_tsunami.py	~15	warn_tsunami_events.py	Tohoku 0.970, wave height norm
test_warn_flood_surge.py	~15	warn_flood_surge_events.py	Sandy surge 0.549
test_pulse_disaster_exposure.py	~20	build_pulse_disaster_exposure.py	RED=200, compound formula
test_wise_multihazard.py	~15	wise_multihazard_decision.py	Nuclear domain, compound escalation
test_warn_hurricane.py	~15	warn_hurricane_events.py	Andrew 0.814, wind/surge norms
waifinders_world tests	73	16 test files	GATE-009 CLOSED
TOTAL	260	All modules	All passing as of 2026-08-09

K. OUTPUT FILE MANIFEST (disaster_demo/)

File	Bytes	Key content	BLAKE3 (first 16)
warn_nuclear_plants.json	23,202	67 plants, NRC live, Indian Point 0.871	20e748a2b77257a6
warn_bc_wildfire_nations.json	24,393	58 nations, Kwadacha 0.7246	323e97c451c4c9c0
warn_tsunami_events.json	642,943	Full DART event archive, Tohoku 0.970	—
warn_hurricane_events.json	36,439	125 Cat-3+ storms, Andrew 0.814	—
warn_earthquake_events.json	~8,000	USGS events, Cascadia 0.748	—
warn_flood_surge_events.json	~6,000	Sandy 0.549	—
pulse_disaster_exposure.json	5,999	2000 bridges, RED=200, NY hazard zones	—
wise_multihazard_decision.json	1,995	EMERGENCY_RESPONSE, compound=True, 6/7 elev	—
warn_earthquake_analogue_1700.json	—	1700 Cascadia analogue	—

L. WAIFINDERS WORLD INTEGRATION

Directory	Status before Aug sprint	Status after sprint
04_PULSE/	EMPTY	PULSE_MULTHAZARD_DOMAIN_REGISTRATION.md
05_WARN/	EMPTY	WARN_MULTHAZARD_DOMAIN_REGISTRATION.md (8 domains)
06_WISE/	EMPTY, TRL 3	WISE_ENGINE_REGISTRATION.md, TRL 5
outputs/disasters/	NOT PRESENT	9 BLAKE3-hashed JSON outputs staged
dashboard/main.py	16 tabs, 3 dead	18 tabs, all live
reports/WAIFINDERS_WORLD_READINESS_MATRIX_WISE_DESIGN	Updated: all TRL bumps, BC row added	

Dashboard tab index after sprint:

Tab #	Name	Data source	Was
0	World Overview	readiness_matrix.csv + WARN metrics	Live
1	Doctrine	00_DOCTRINE/	Live
2	SENTINEL	03_SENTINEL/	Live
3	PULSE	pulse_disaster_exposure.json	Stub
4	WARN	wise_multihazard_decision.json	MD only
5	WISE	wise_multihazard_decision.json	Design phase
6	RELATE	—	Placeholder
7	S.A.F.E.	08_SAFE_AUDIT_EVIDENCE/	Live
8	Wildfire Validation	replay_validation/	Live
9	BC Wildfire Nations	warn_bc_wildfire_nations.json	NEW
10	INDG BC Nations SPARK7	warn_spark7_bridge (live compute)	NEW
11	Water Validation	GATE-004 CLOSED	■ Design
12	Digital Twins	CONCEPT	CONCEPT
13	Evidence Ledger	BLAKE3 manifests	Live
14	Funding Readiness	FUNDING_READY_SUMMARY.md	Live
15	Partner Packages	PARTNER_READY_SUMMARY.md	Live
16	Claim Control	CLAIMS_REGISTER.md	Live

M. READINESS MATRIX — CURRENT STATE

Component	Status	TRL	Demo	Pilot	Enterprise	Prod
WARN	PILOT-READY	6	YES	YES	NO	NO

Component	Status	TRL	Demo	Pilot	Enterprise	Prod
PULSE	PILOT-READY	6	YES	YES	NO	NO
WISE	PILOT-READY	5	YES	YES	NO	NO
SENTINEL	PILOT-READY	5	YES	YES	NO	NO
S.A.F.E.	PILOT-READY	5	YES	YES	NO	NO
SPARK7 (BC)	PILOT-READY	5	YES	YES	NO	NO
BC Wildfire Nations	PILOT-READY	5	YES	YES	NO	NO
Water Infrastructure	GATE-004-CLOSED	5	YES	YES	NO	NO
Demo Dashboard	LIVE	—	YES	YES	NO	NO
Bow River DT	CONCEPT	1	NO	NO	NO	NO
RELATE	FRAMEWORK	3	NO	NO	NO	NO

N. CLAIM BOUNDARY — APPROVED AND PROHIBITED

APPROVED (with stated qualifiers):

- ✓ WARN is a replay-tested multi-hazard risk-prioritization engine [Alberta replay real data; external review pending]
- ✓ WARN BC provides wildfire risk proximity scores for First Nations using public BC data [approximate centroids; Nations decide]
- ✓ PULSE provides multi-hazard infrastructure stress indicators [county-level; not structural engineering]
- ✓ WISE synthesises multi-hazard signals into operational decision states [research prototype; not validated for emergency management]
- ✓ SPARK7 projects 10-year community readiness under compound climate pressure [preliminary baselines; no Nation consent yet]
- ✓ GATE-004 CLOSED — water provenance confirmed (Toronto, Calgary, Kitchener OGL)
- ✓ BLAKE3 is the sole hashing standard — no SHA-256, no MD5

PROHIBITED — must not appear in any external material:

- ✗ WARN predicts wildfires (or any hazard outcome)
- ✗ ROC-AUC 0.998 — superseded (label leakage, synthetic 500-event set)
- ✗ Alberta WARN performance transfers to BC or Ontario without separate replay
- ✗ BC Wildfire Service endorses WAIFINDERS
- ✗ Any Nation endorses WAIFINDERS
- ✗ Nation cultural data was used without consent
- ✗ WISE/PULSE autonomously directs emergency response
- ✗ SENTINEL adopted by federal government
- ✗ Calgary / Toronto / Kitchener endorse WAIFINDERS

O. KEY FILE LOCATIONS

File / module	Repository path
WARN Alberta engine	waifinders_wildfire/scripts/
WARN BC Nations engine	waifinders_disasters/scripts/build_warn_bc_wildfire_nations.py
WARN Nuclear engine	waifinders_disasters/scripts/build_warn_nuclear.py
WARN Earthquake engine	waifinders_disasters/scripts/build_warn_earthquake_events.py
WARN Tsunami engine	waifinders_disasters/scripts/build_warn_tsunami_events.py
WARN Hurricane engine	waifinders_disasters/scripts/build_warn_hurricane_events.py
WARN Flood Surge engine	waifinders_disasters/scripts/build_warn_flood_surge_events.py
PULSE engine	waifinders_disasters/scripts/build_pulse_disaster_exposure.py
WISE engine	waifinders_disasters/scripts/wise_multihazard_decision.py
SPARK7 module	waifinders_wildfire/modules/spark7.py
SPARK7 BC Nations JSON	waifinders_wildfire/data/spark7/spark7_bc_nations.json
WARN→SPARK7 bridge	waifinders_wildfire/modules/warn_spark7_bridge.py
INDG BC Nations dashboard	waifinders_wildfire/pages/09_INDG_BC_Nations.py

File / module	Repository path
WAIFINDERS World dashboard	WAIFINDERS_WORLD/dashboard/main.py
Readiness matrix	WAIFINDERS_WORLD/reports/WAIFINDERS_WORLD_READINESS_MATRIX.csv
Disaster outputs	waifinders_disasters/outputs/disaster_demo/*.json
World disaster outputs copy	WAIFINDERS_WORLD/outputs/disasters/*.json
All tests (disasters)	waifinders_disasters/tests/test_*.py
All tests (wildfire)	waifinders_wildfire/tests/test_*.py
Plain language PDF	waifinders_disasters/materials/waifinders_plain_language_capabilities.pdf
This PDF	waifinders_disasters/materials/waifinders_full_build_metrics.pdf
Completion status PDF	waifinders_disasters/materials/waifinders_world_completion_status.pdf

P. NEXT STEPS — PRIORITY ORDER

Priority	Action	Blocks	Who
1 — CRITICAL	Send external peer review invitations (REVIEWER_OUTREACH_EMAIL.md ready)	Production WARN claim	WAIFINDERS → reviewers
2 — CRITICAL	Send Alberta wildfire data request letter (REQUEST_LETTER.md ready)	GATE-001	WAIFINDERS → Alberta
3 — CRITICAL	Send NRCan + Public Safety Canada SENTINEL package (GATE-006 ready)	Federal deployment	WAIFINDERS → NRCan
4 — HIGH	Nation-to-Nation engagement for BC first data agreement	BC WARN production	WAIFINDERS → Nation
5 — HIGH	BC Wildfire Management Branch data agreement	BC WARN production	WAIFINDERS → BC
6 — HIGH	City operational data agreement (Toronto or Calgary)	PULSE production	WAIFINDERS → City
7 — MEDIUM	Ontario wildfire replay (separate run required)	Ontario WARN	Requires ON data
8 — MEDIUM	Bow River Digital Twin GIS + Treaty 7 engagement	Digital Twins	WAIFINDERS + Treaty 7